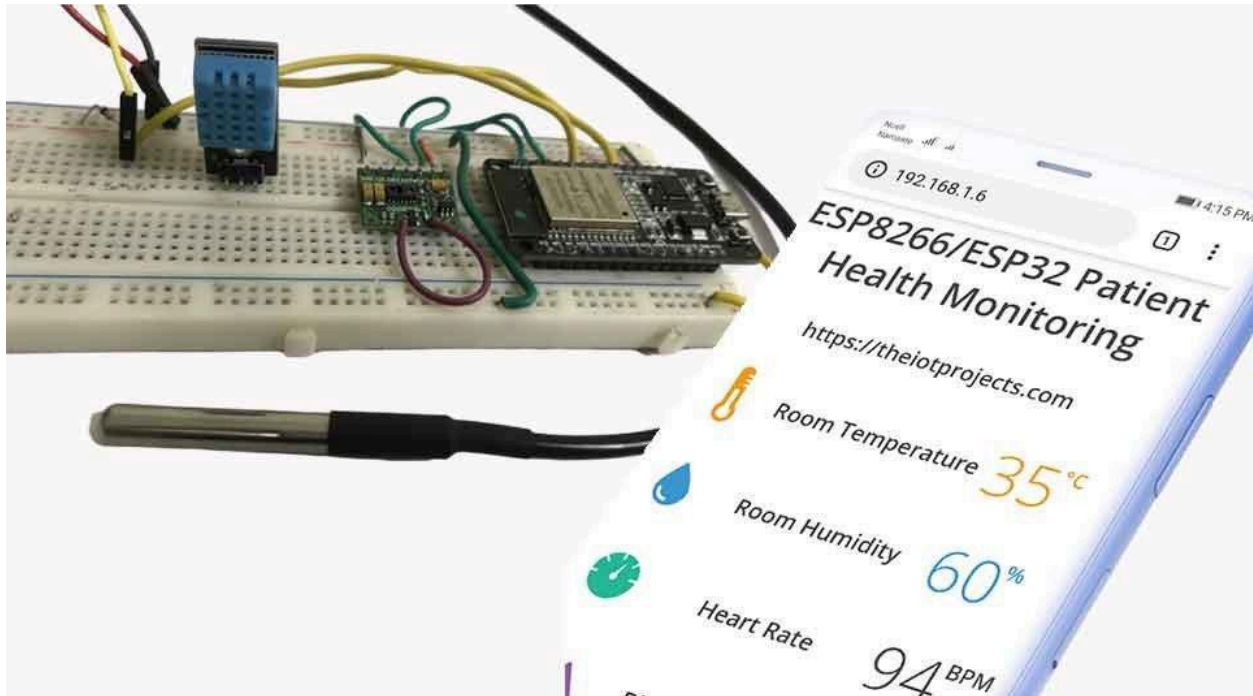


IoT (Internet of Things)

IoT (Internet of Things)











What is IoT?

IoT means connecting physical devices to the internet so they can:

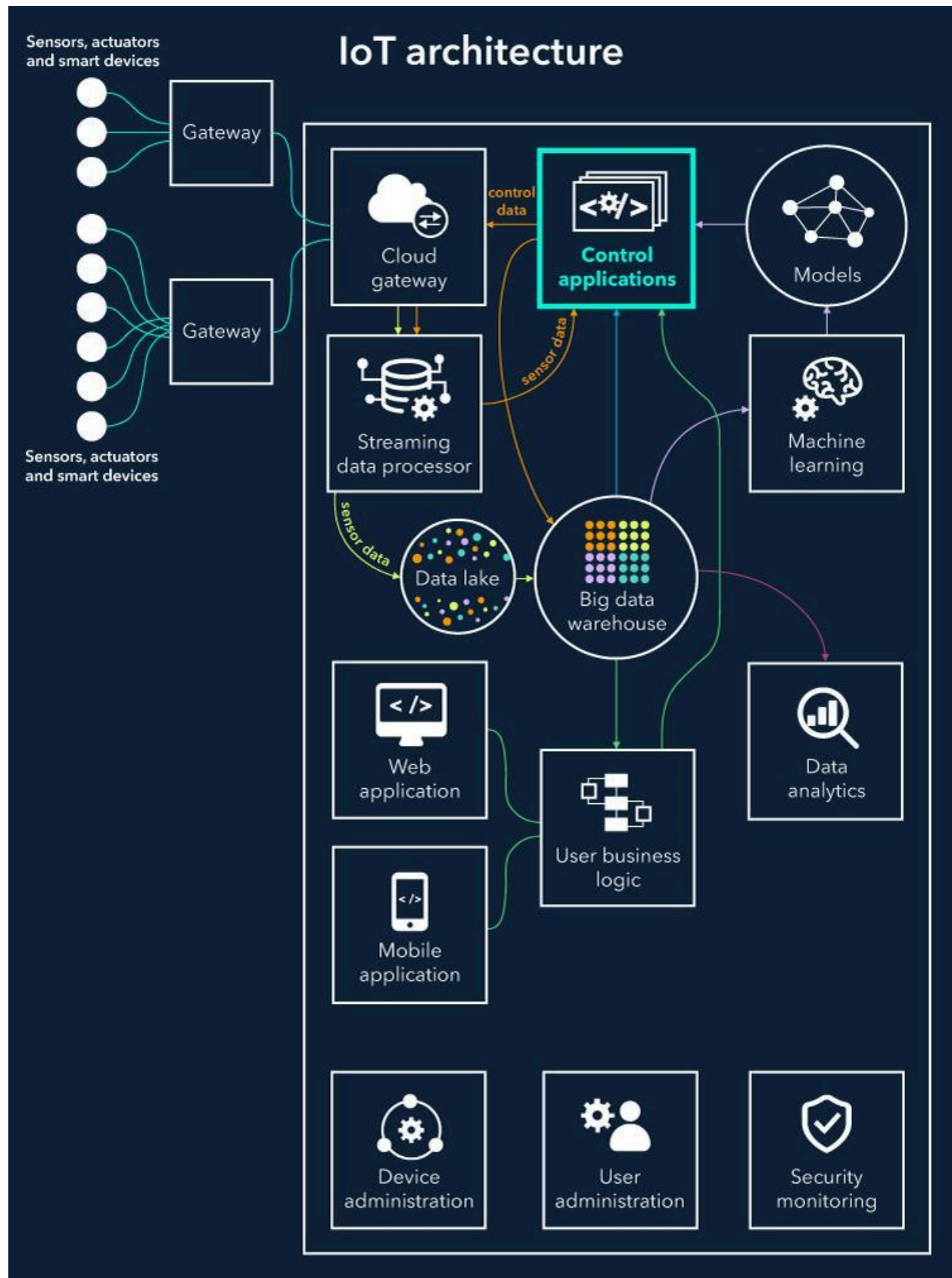
- Collect data
- Send data
- Receive commands
- Work automatically

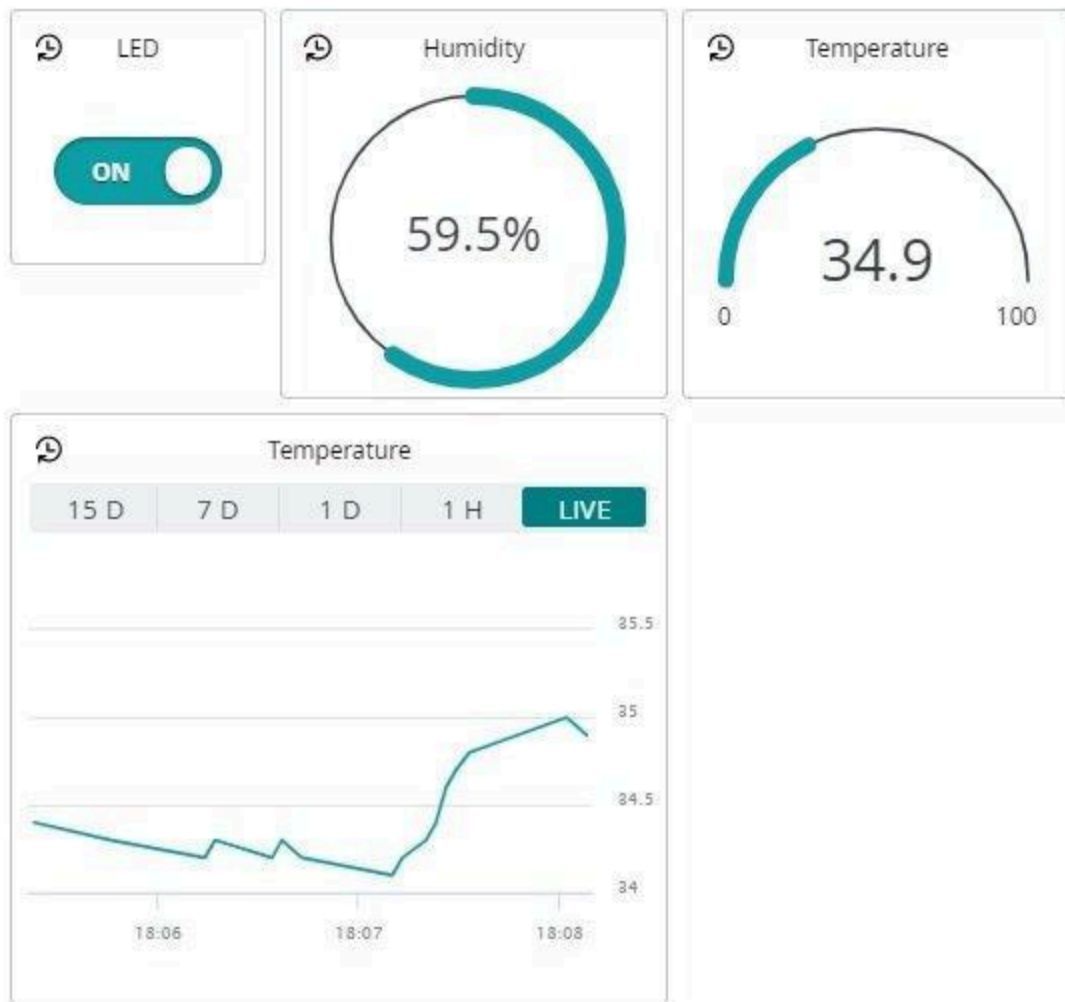
Example:

- Smart lights
- Smart agriculture
- Smart CCTV
- Smart factories

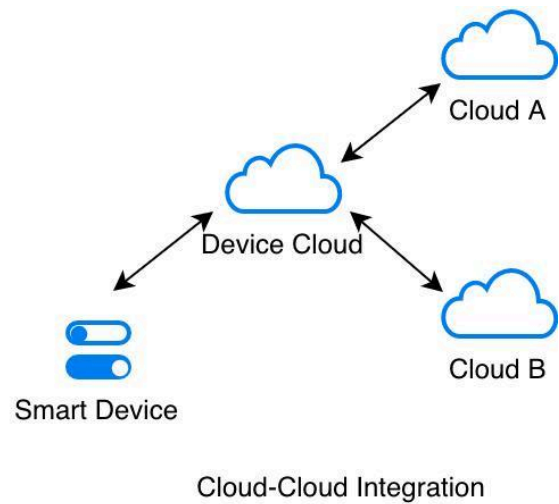
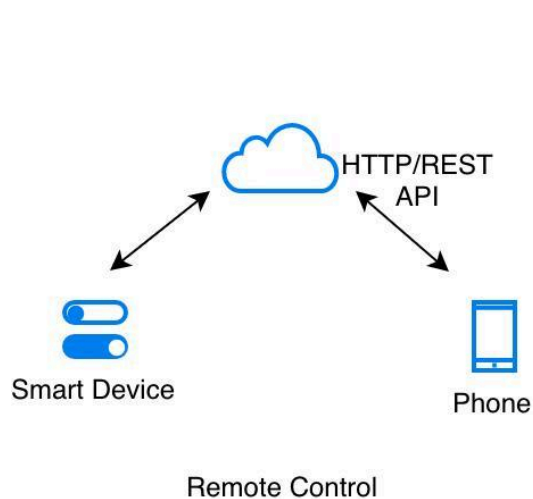
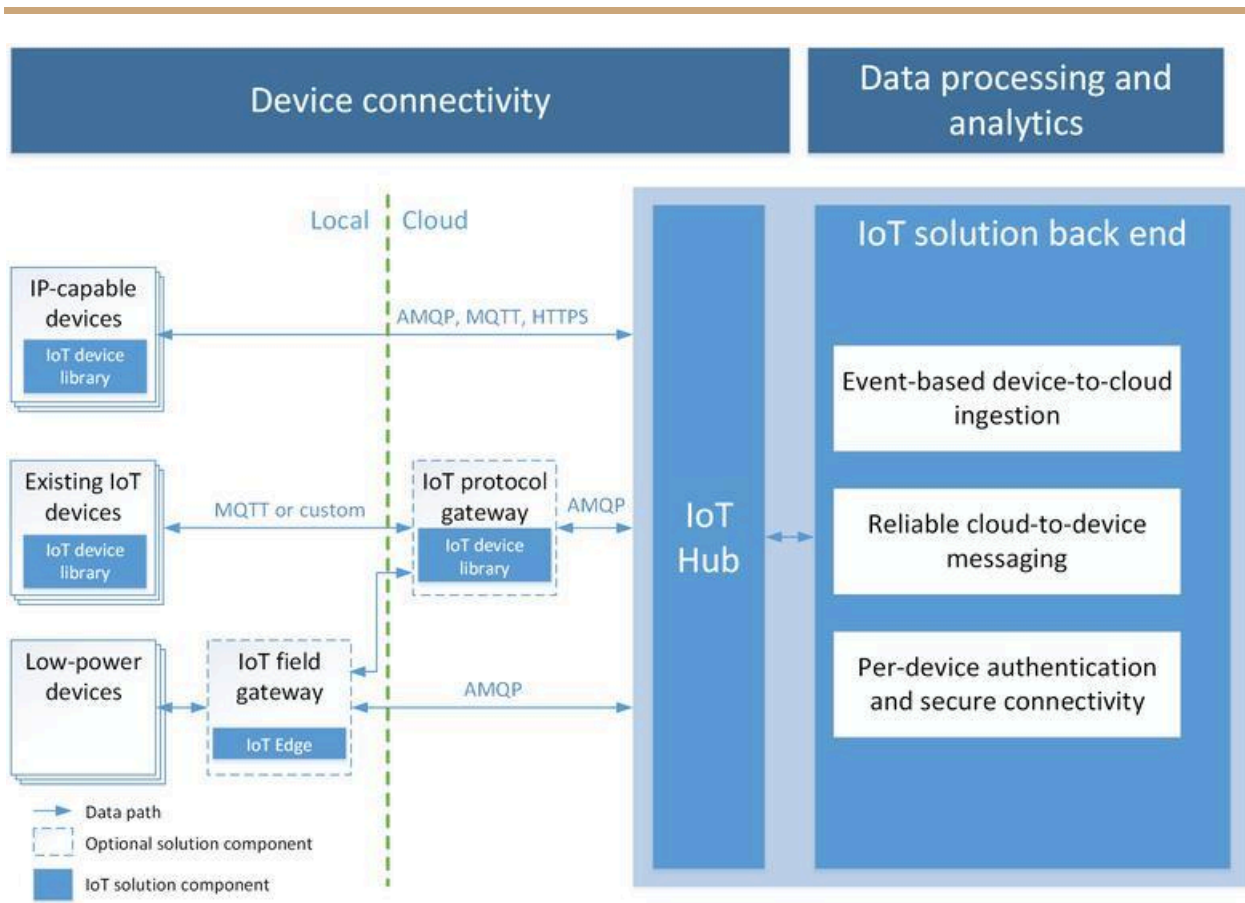
-
- Health monitoring devices
-

Basic IoT Architecture













Flow

Sensor → Microcontroller → Internet → Cloud → Mobile App

Example:

1. Temperature sensor reads data

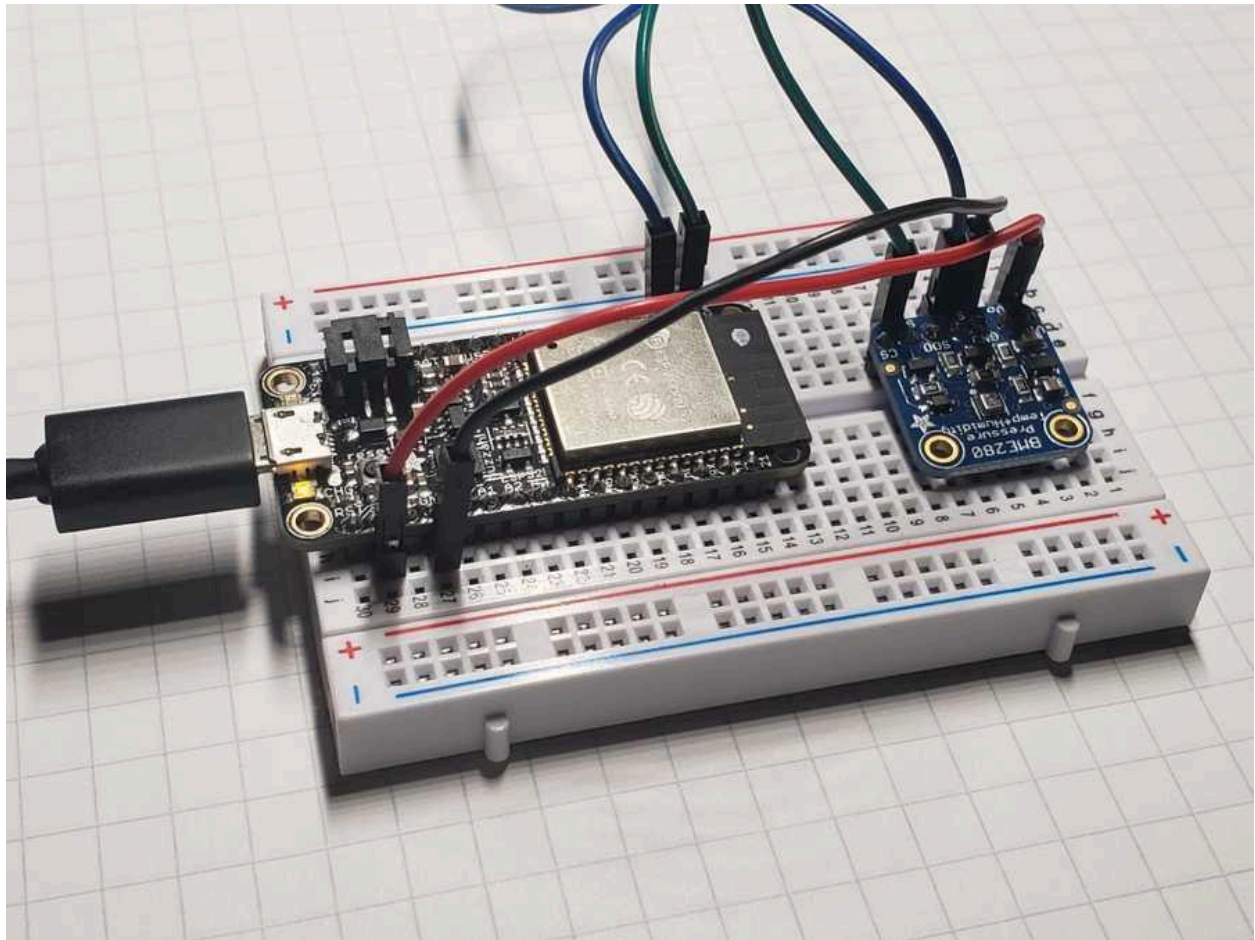
-
2. ESP32 processes data
 3. Wi-Fi sends data to cloud
 4. Mobile app shows live status
-

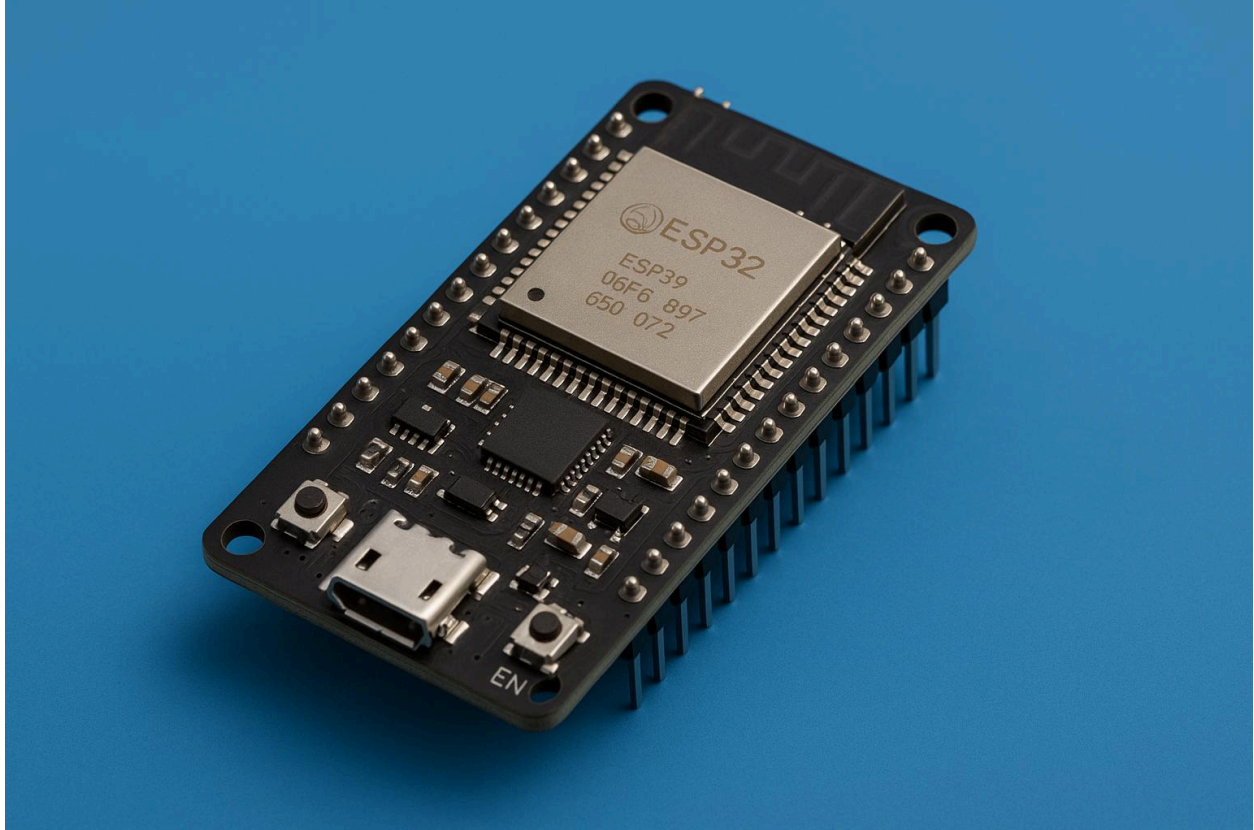
Main IoT Components

Component	Purpose
ESP32	Main controller
Sensors	Data collection
Wi-Fi/LoRa	Communication
Cloud Platform	Data storage
Mobile App	User control
Relay	Device switching

Popular IoT Hardware

ESP32



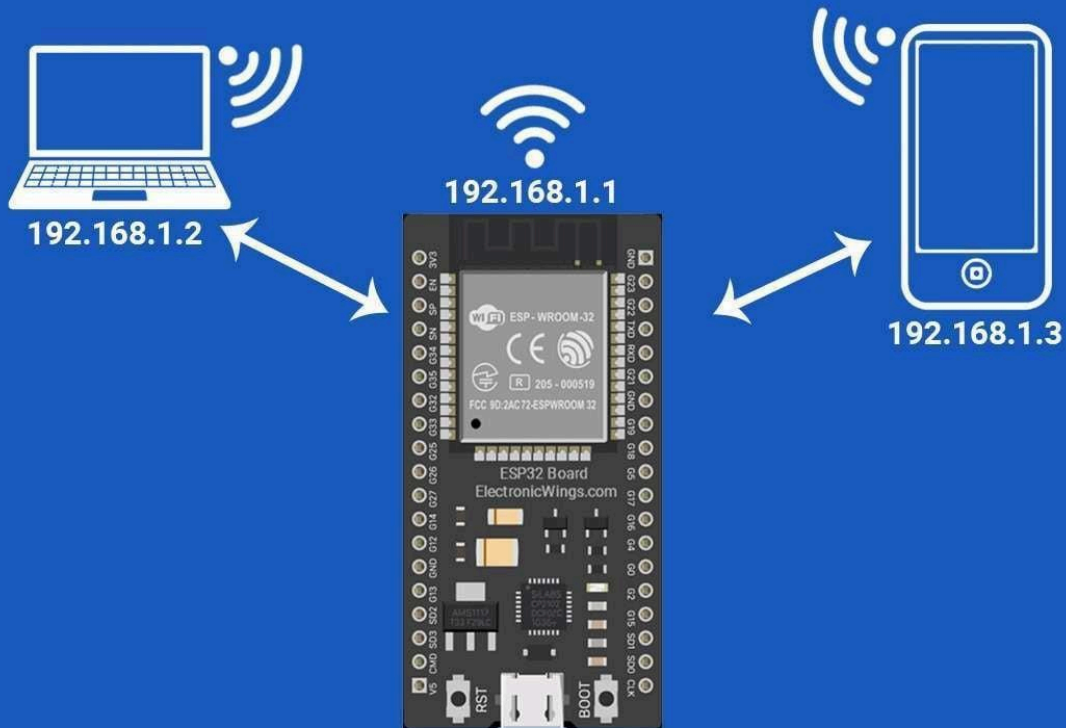








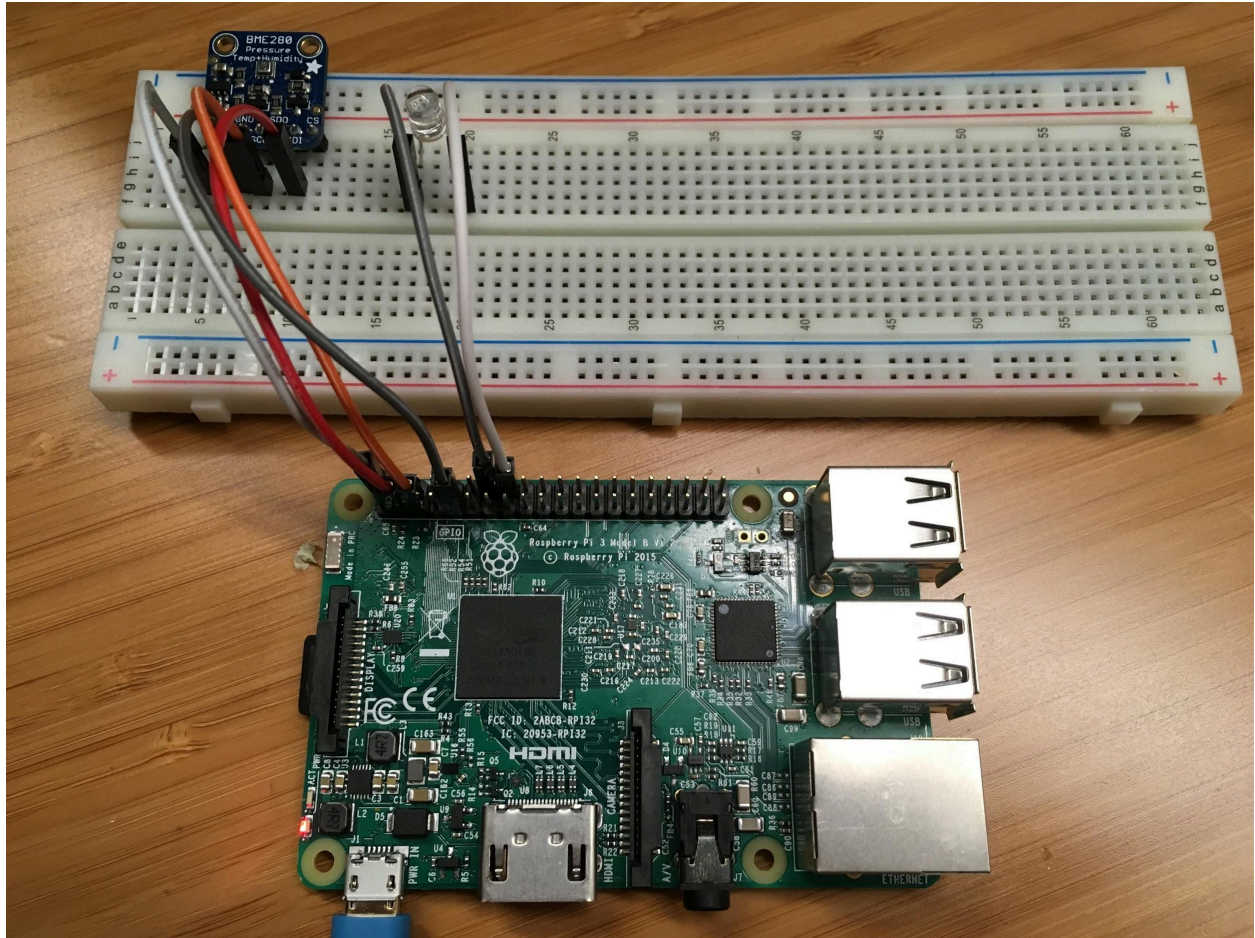
How to create ESP32 WiFi Server

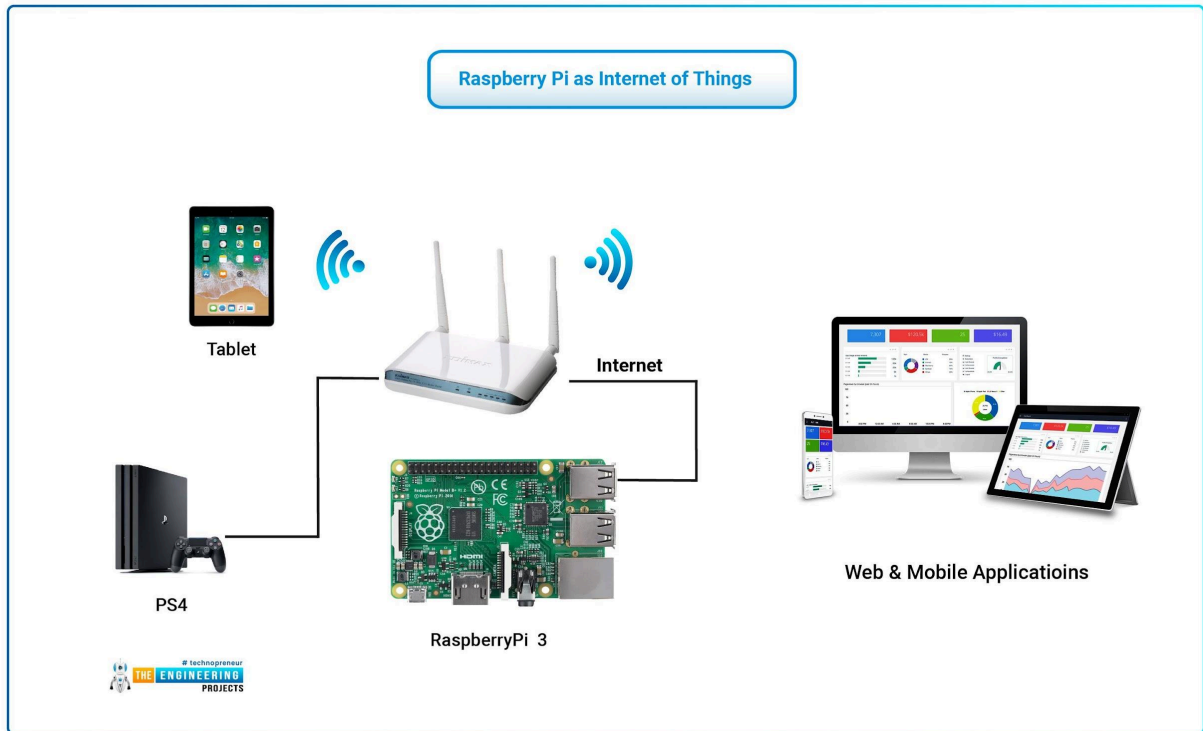


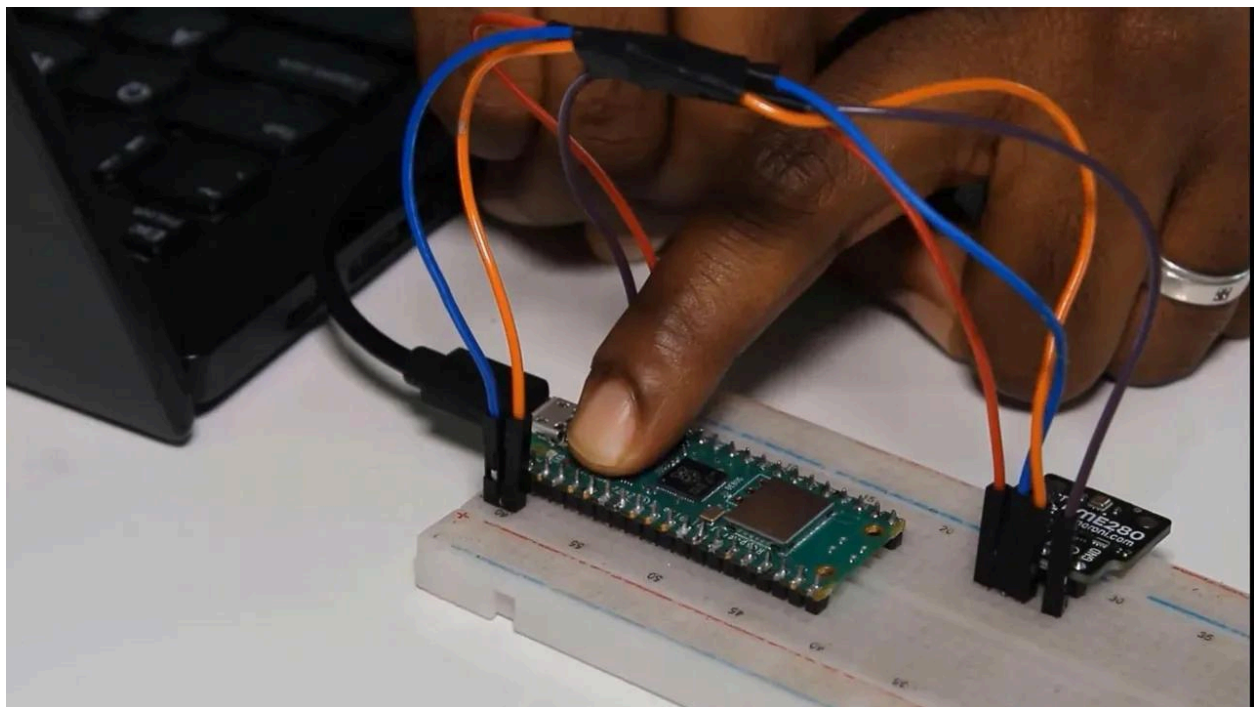
Features

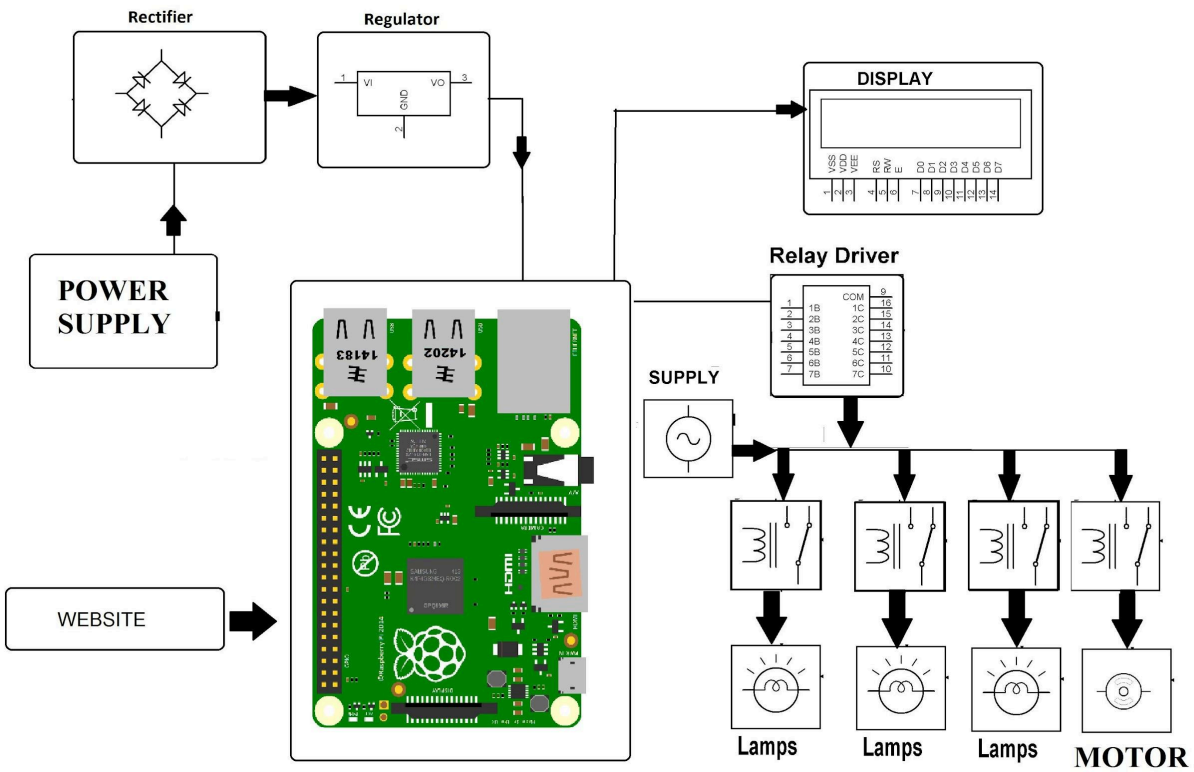
- Wi-Fi
- Bluetooth
- Low power
- Cheap and powerful

Raspberry Pi







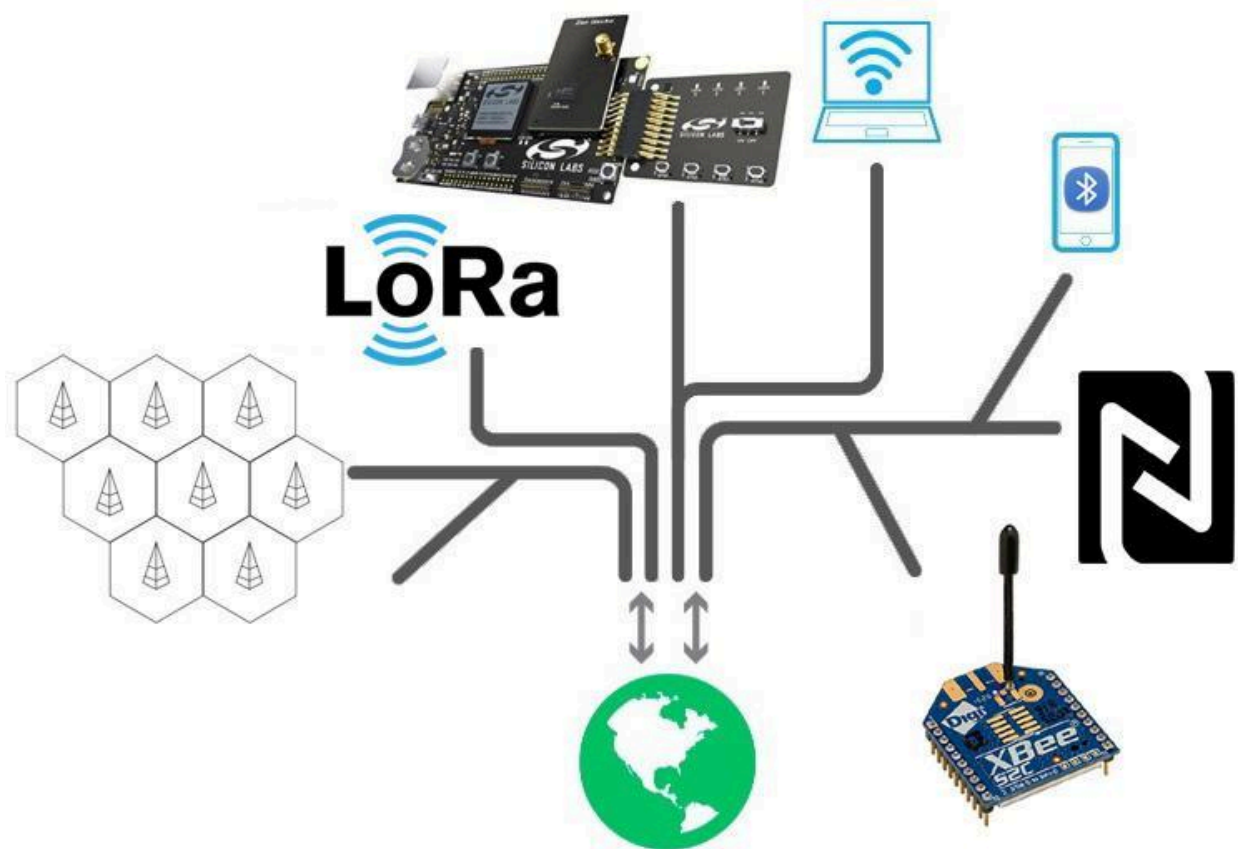


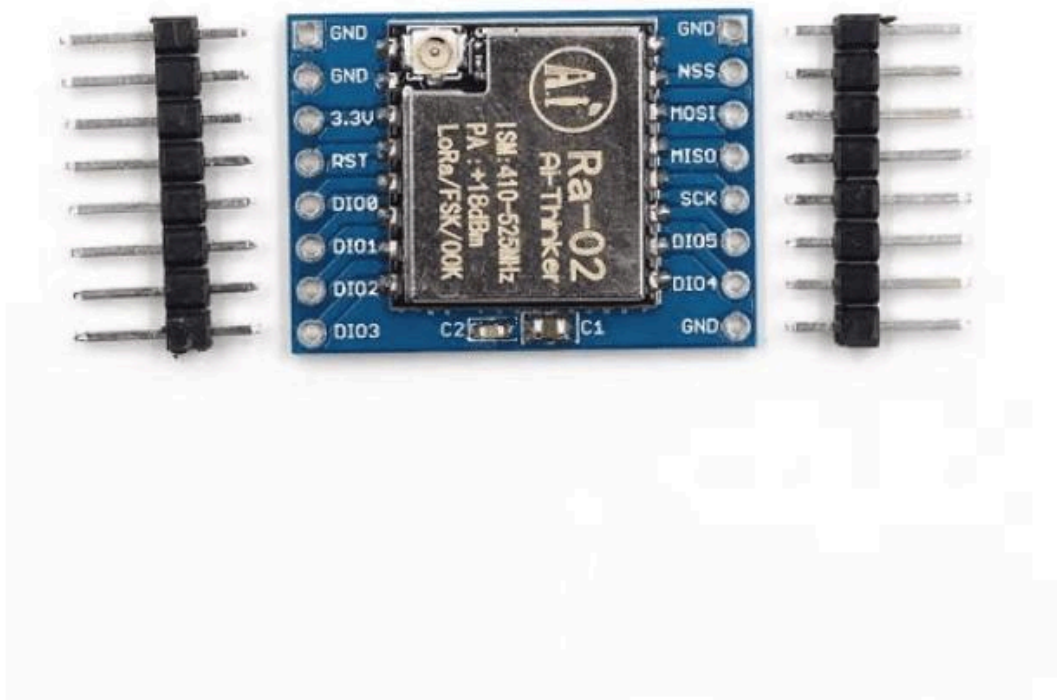
Best For

- AI + IoT
- CCTV
- Servers
- Dashboards

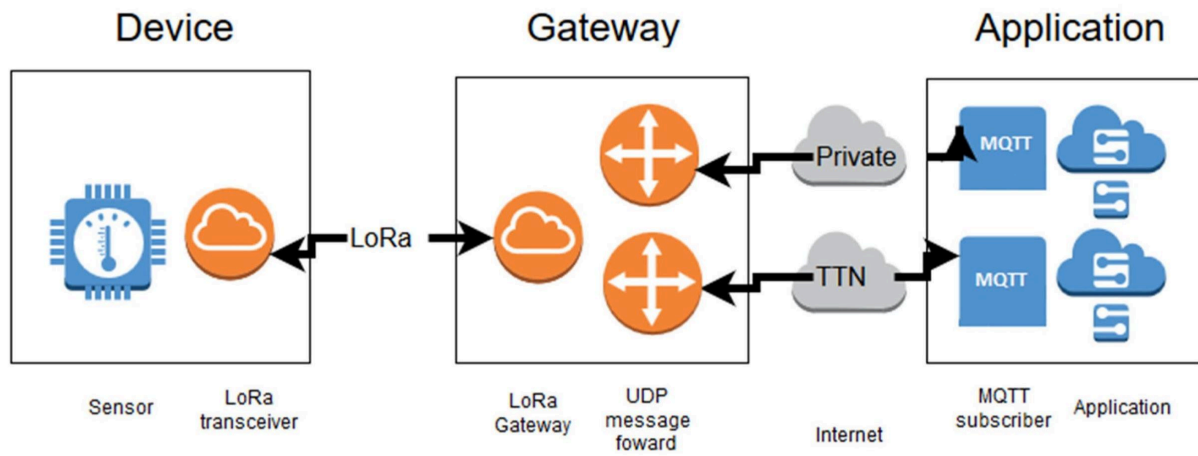
LoRa Modules







LoRa Architecture





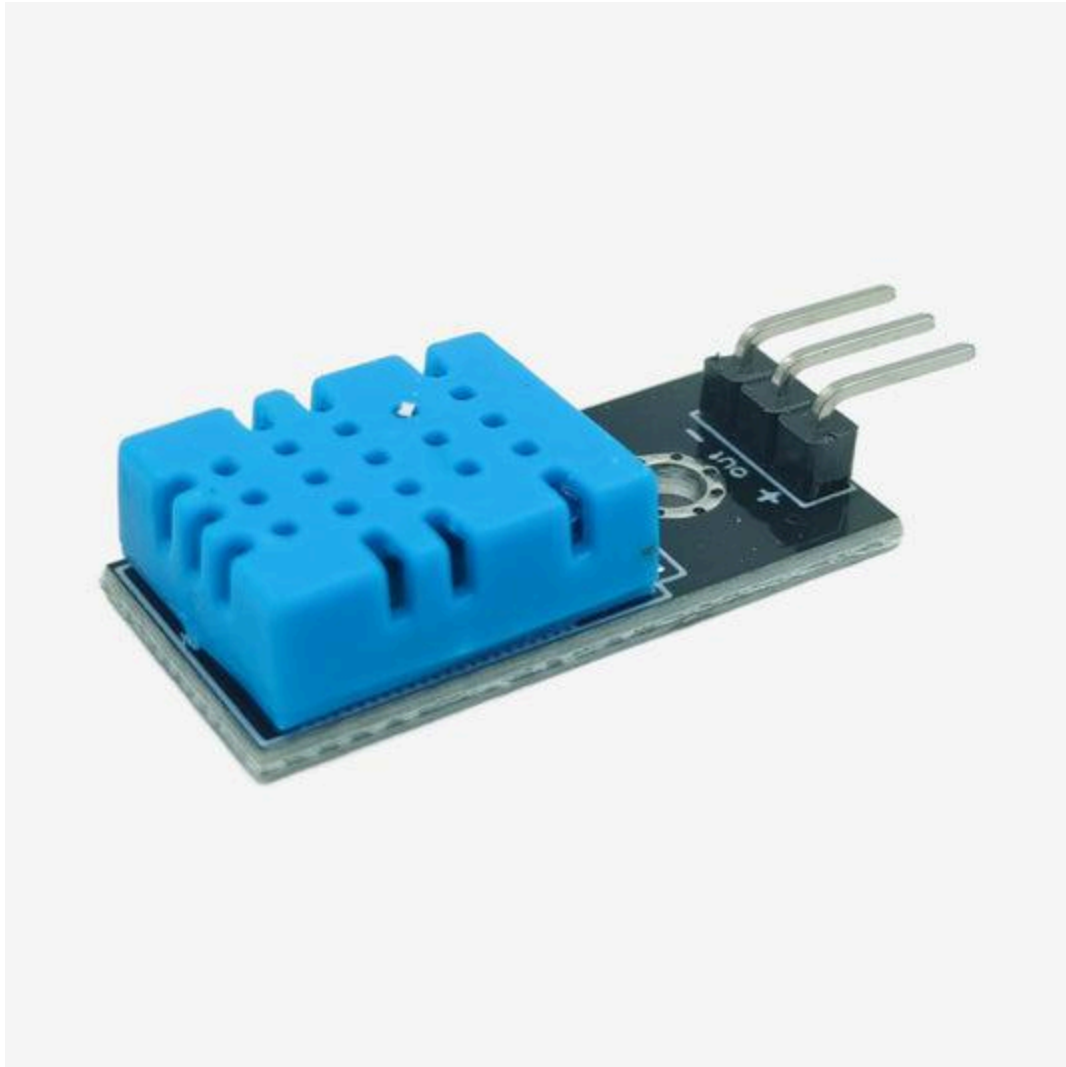
Best For

- Long-range communication

-
- Agriculture
 - Remote monitoring
-

Important IoT Sensors

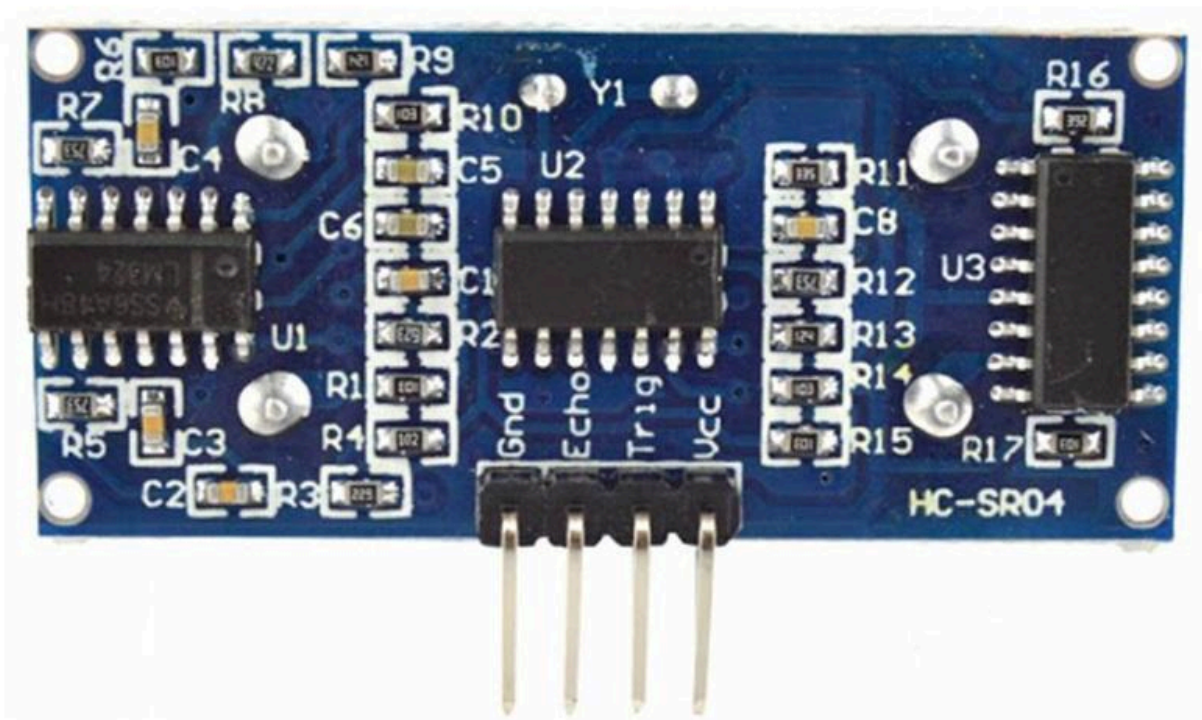












Sensor	Use
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DHT11	Temperature + humidity
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Ultrasonic Distance
 measurement

Soil Moisture Agriculture

MQ Sensors Gas detection

PIR Sensor Motion detection

GPS Tracking

IoT Communication Technologies



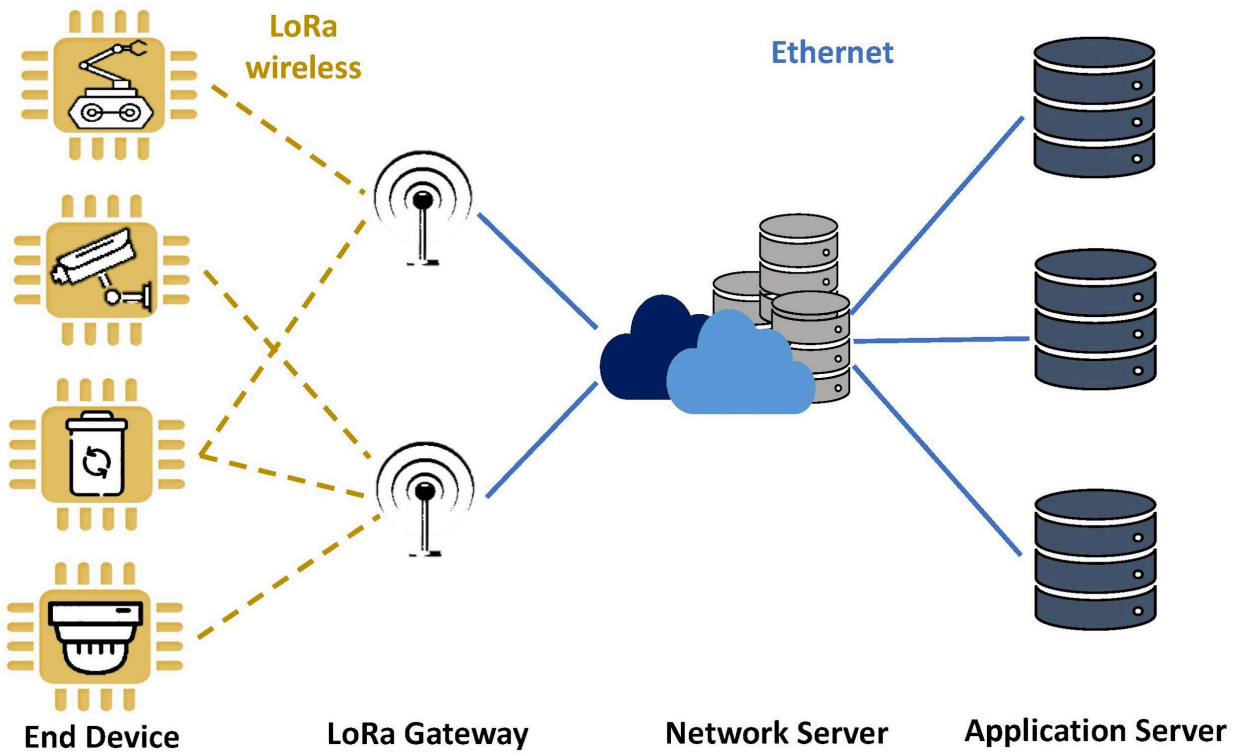


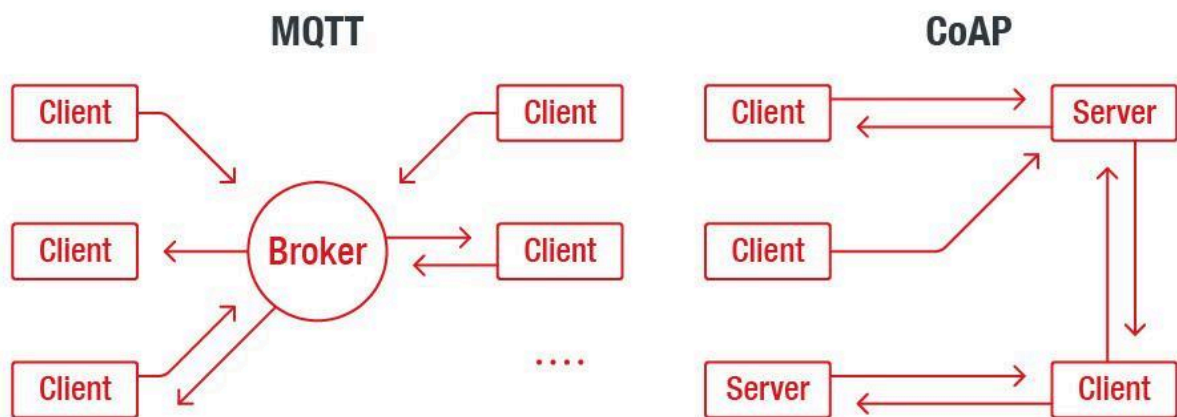
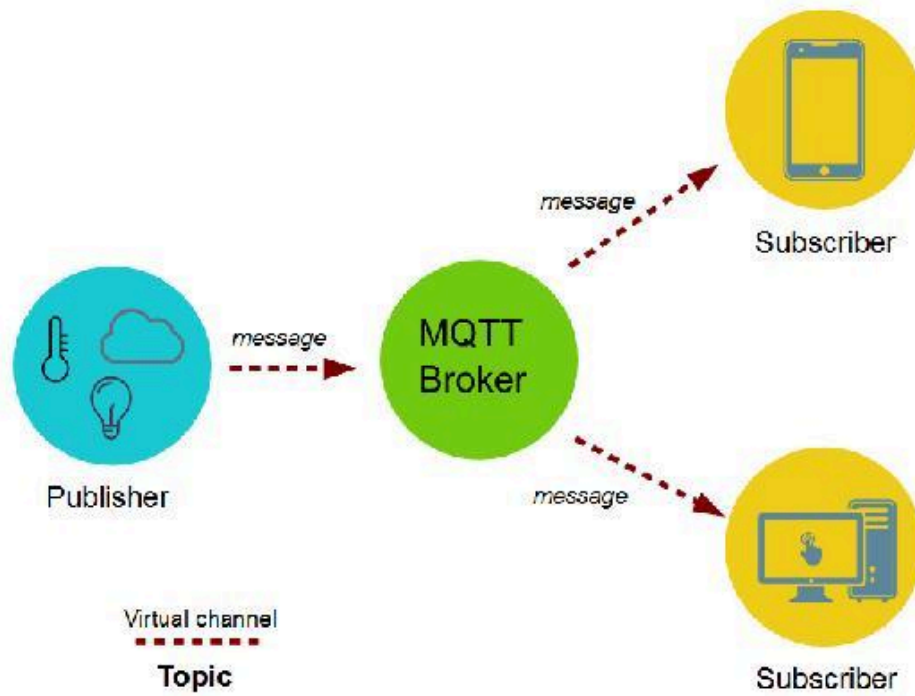
INTRODUCTION TO MQTT PROTOCOL

MQTT stands for Message Queuing Telemetry Transport.



A publish/subscribe messaging protocol for IoT connectivity.





Technology **Range**
y

Wi-Fi	Medium
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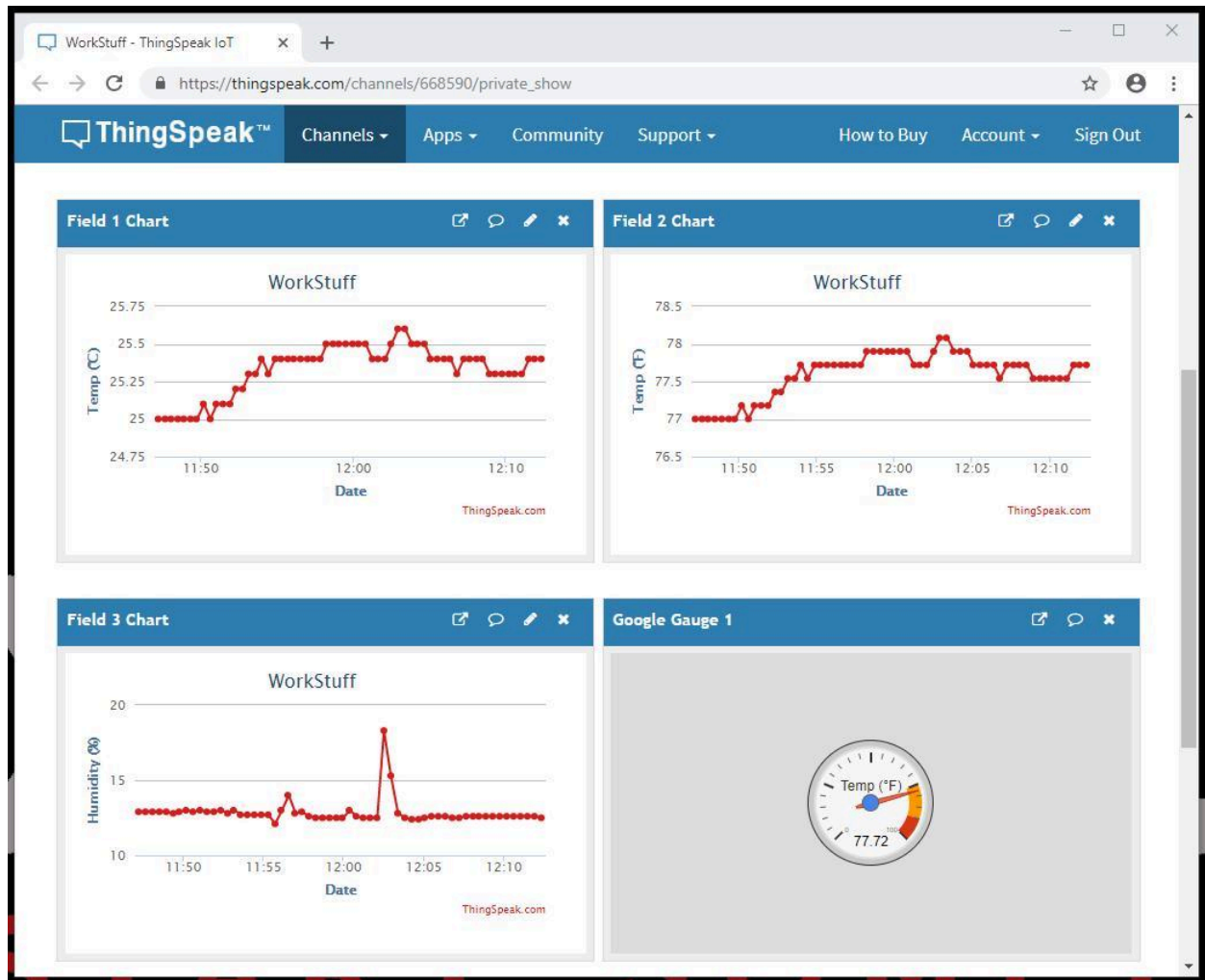
Bluetooth	Short
-----------	-------

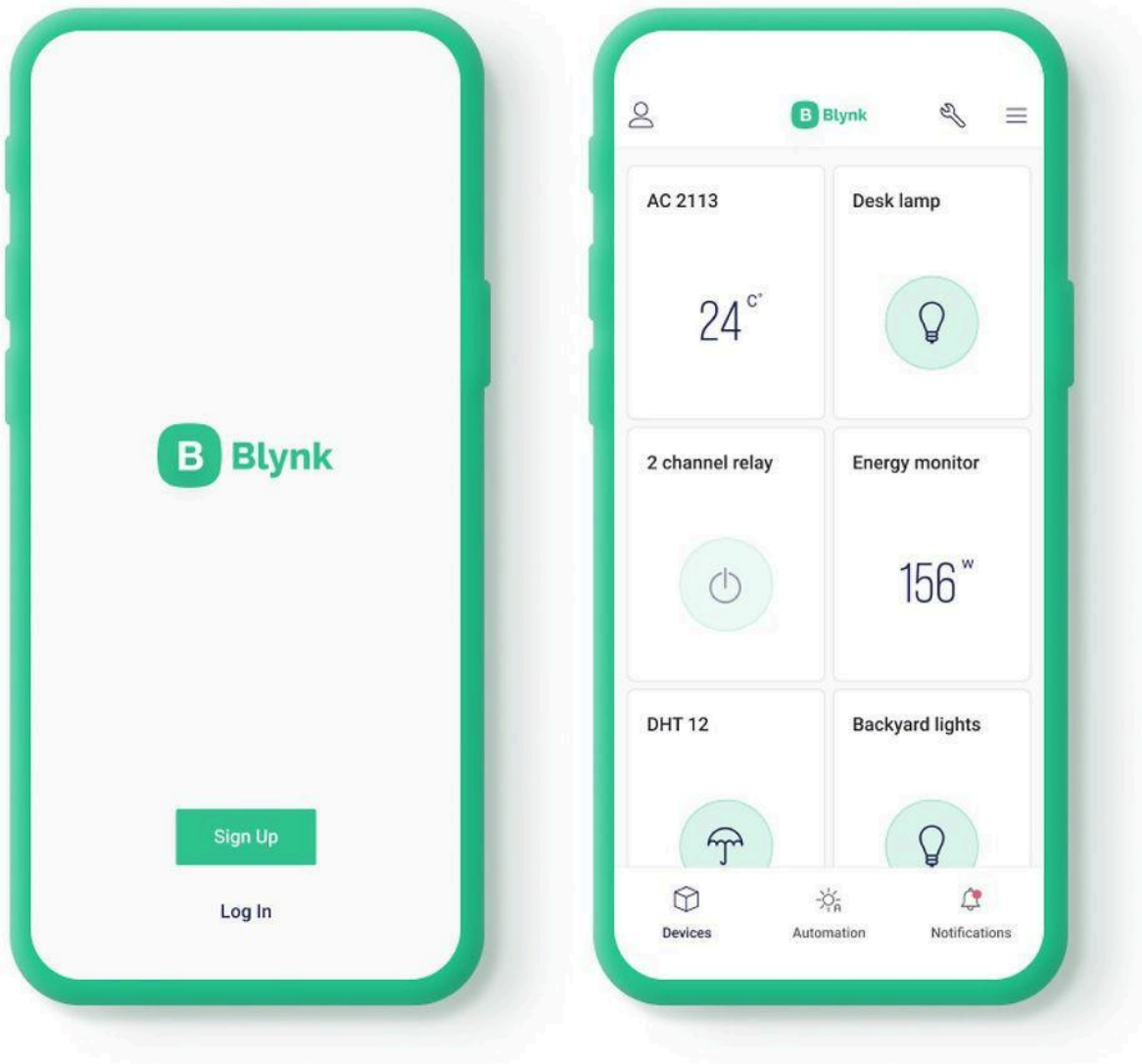
LoRa	Very Long
------	-----------

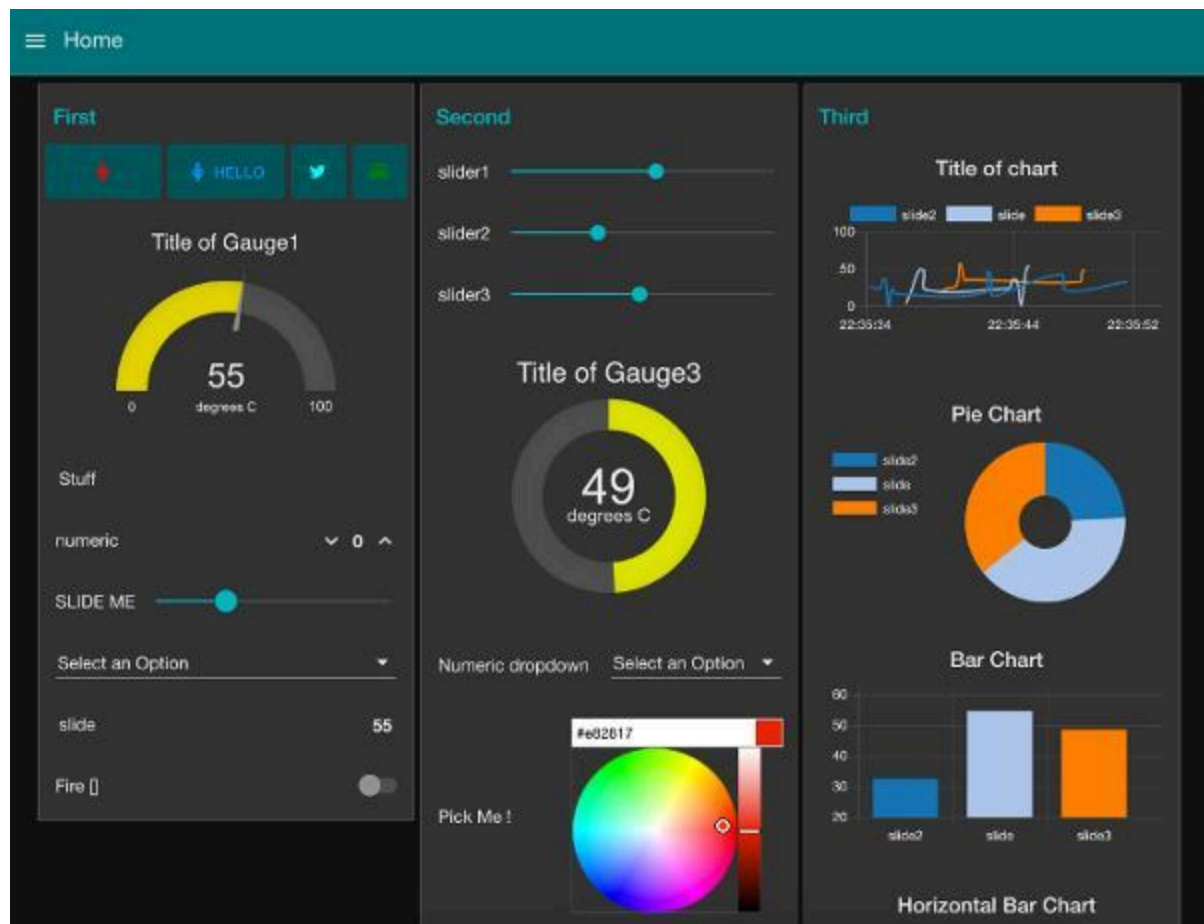
Zigbee	Medium
--------	--------

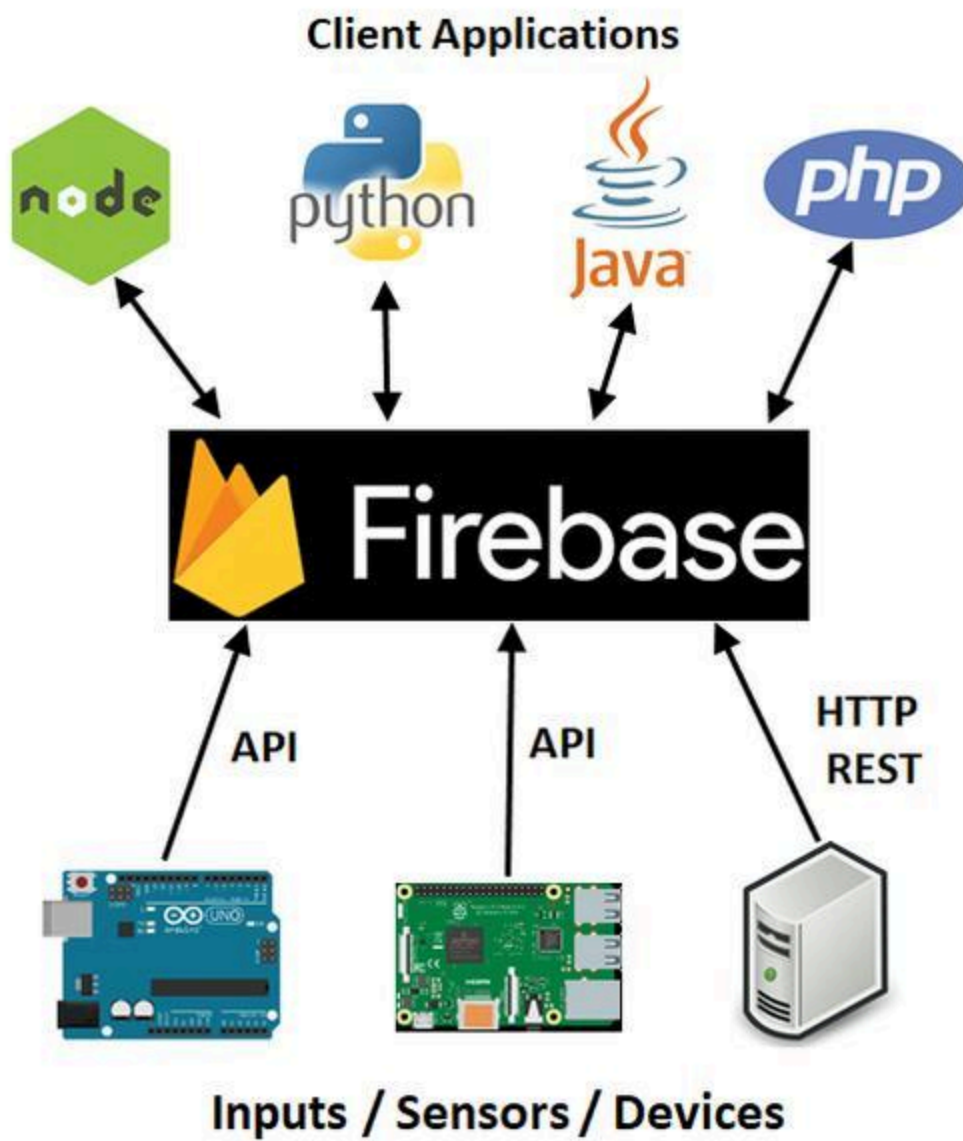
GSM/4G	Very Long
--------	-----------

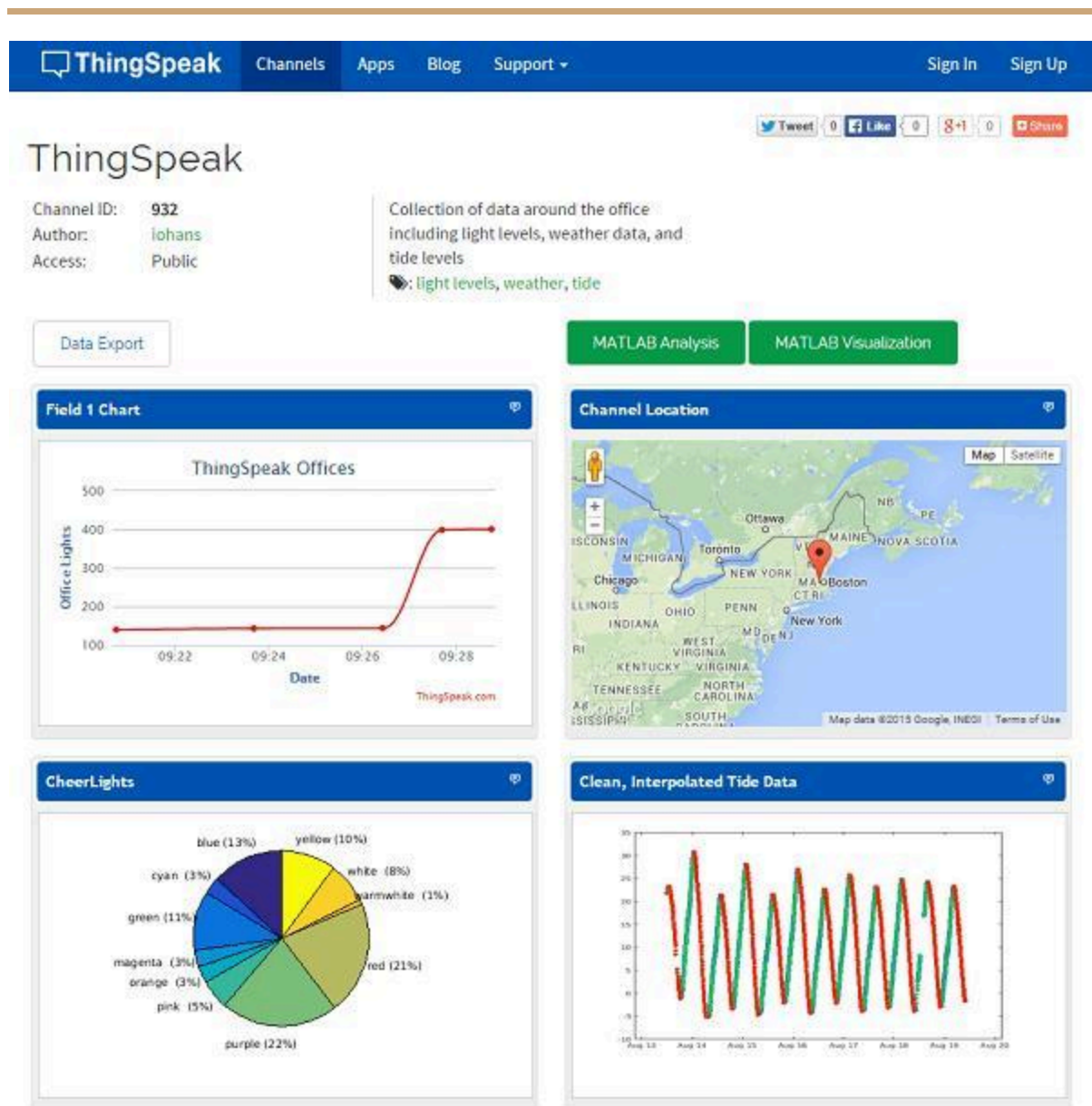
Cloud Platforms for IoT











Platform	Purpose
Blynk	Mobile control
ThingSpeak	Data graphs
Firebase	Real-time database

Node-RED Visual automation

MQTT Broker Device
 communication

Beginner IoT Projects

1. Smart Light Control

- ESP32
- Relay
- Mobile app

2. Weather Monitoring System

- DHT11
- ESP32
- Cloud dashboard

3. Smart Agriculture

- Soil moisture sensor
- Water pump
- Automatic irrigation

4. IoT Home Automation

- Wi-Fi control

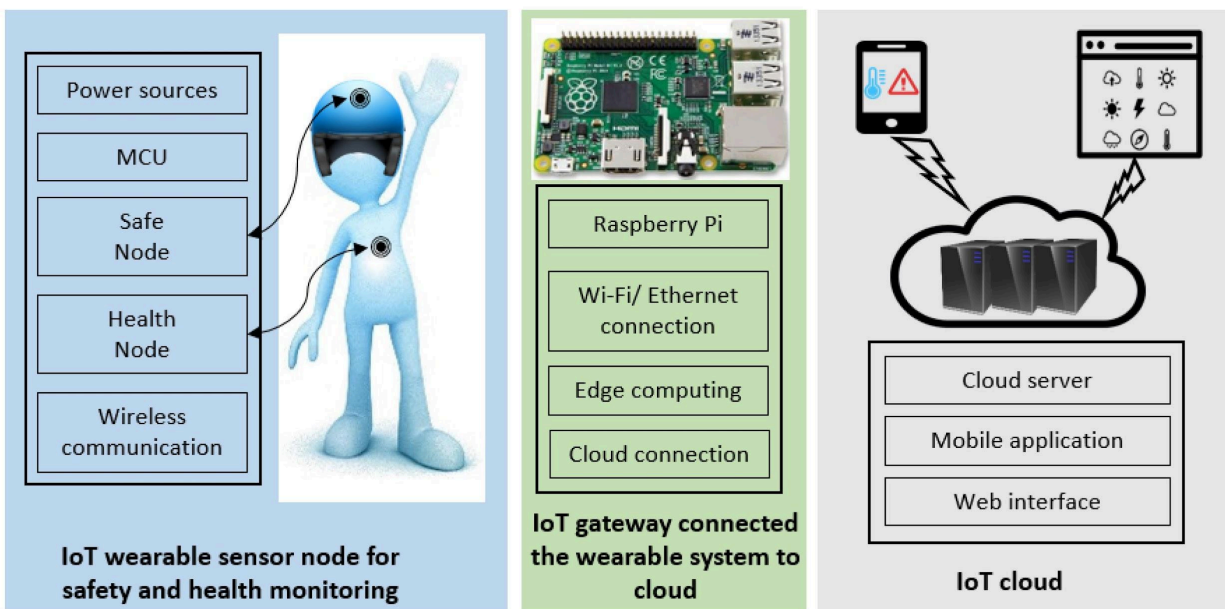
-
- Smart switches
 - Voice commands
-

Intermediate IoT Projects









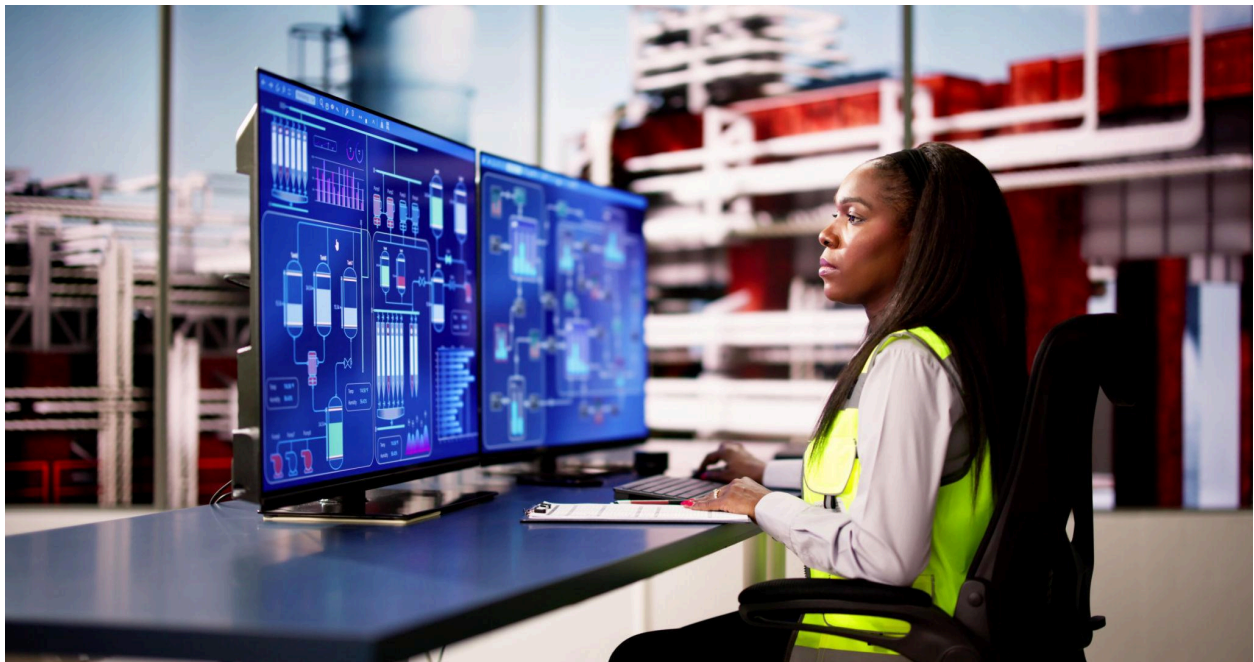


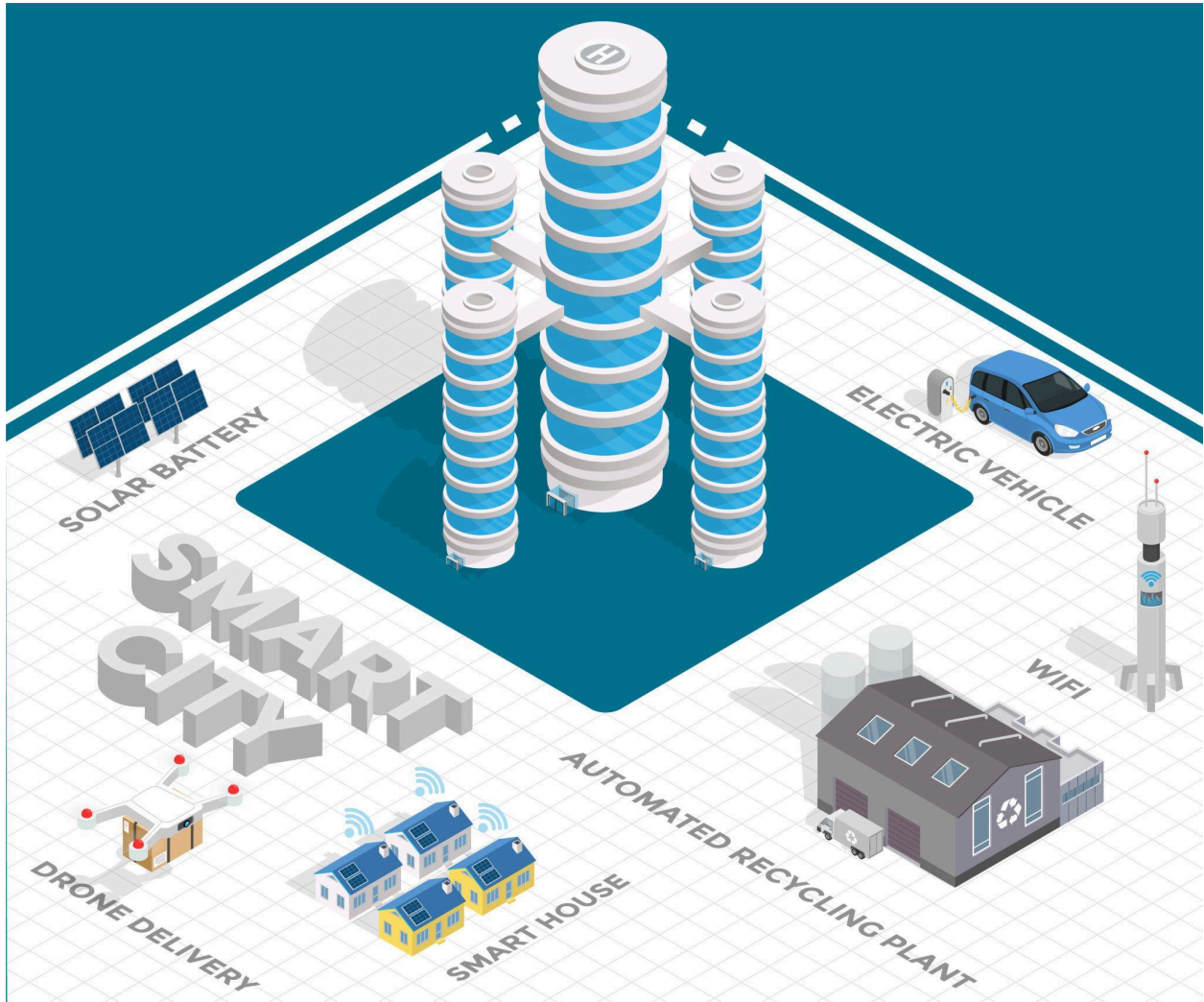
- AI CCTV
- Smart parking
- Smart energy meter
- Vehicle tracking
- Industrial monitoring

Advanced IoT Projects









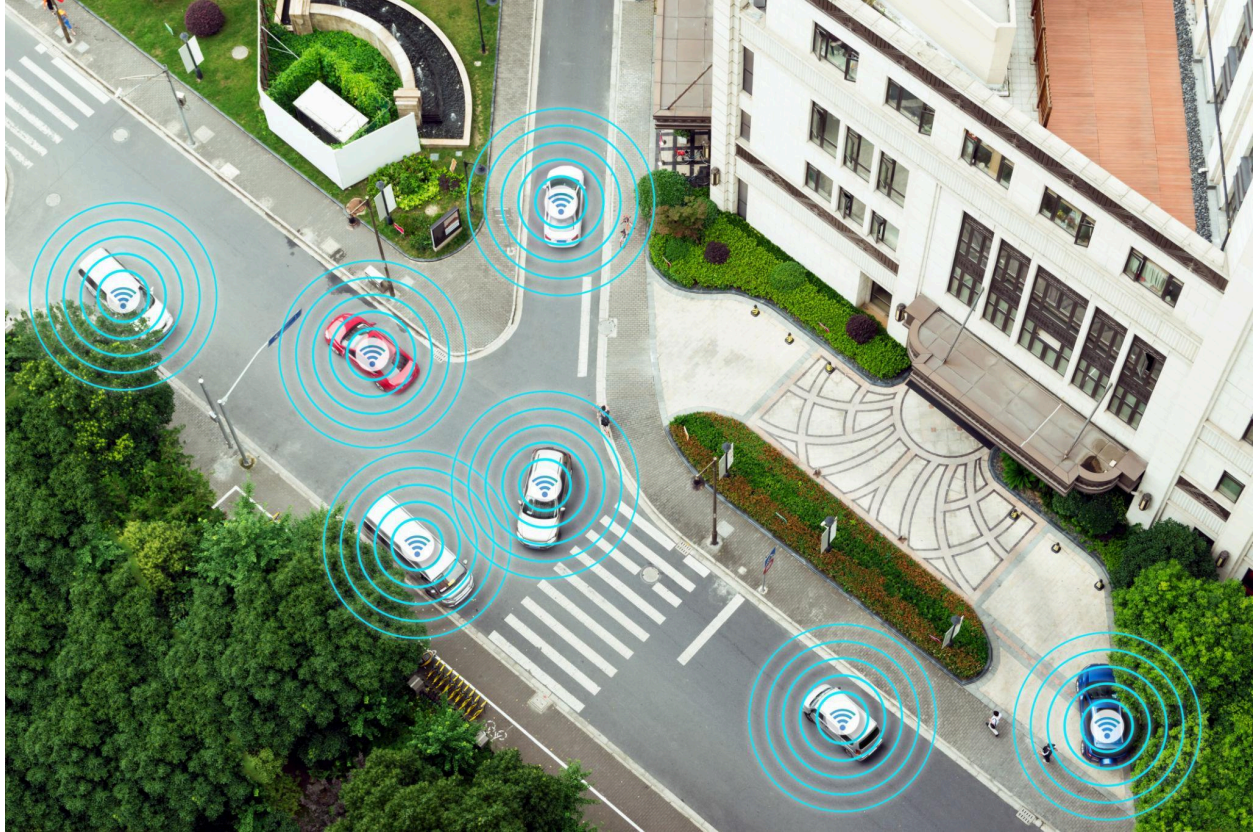




- AI surveillance system
- Drone monitoring
- Smart city system
- Autonomous robot
- Industrial automation

IoT + AI + Robotics

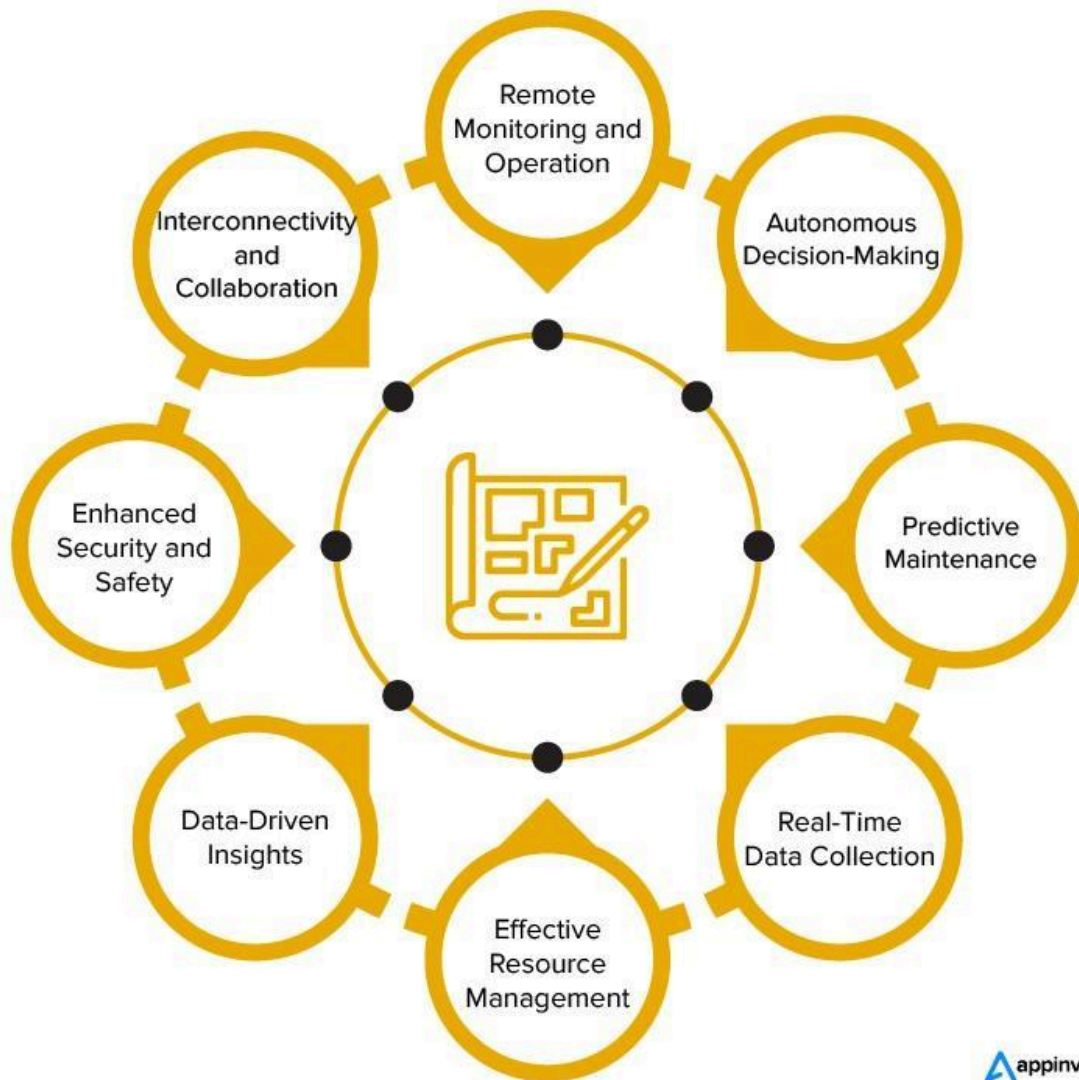








Must-Have Features of IoT in Robotics





This is one of the strongest future technologies.

Used In

- Smart cities
- Defense systems
- Healthcare
- Agriculture
- Aerospace
- Industrial automation

Best Languages for IoT

Language	Purpose
C/C++	Embedded programming
Python	AI + Raspberry Pi
JavaScript	Dashboards
MicroPython	ESP32 scripting

Best IoT Learning Roadmap

1. Electronics basics
2. Arduino + ESP32
3. Sensors interfacing
4. Wi-Fi + MQTT
5. Cloud dashboards
6. Raspberry Pi
7. OpenCV + AI
8. Robotics + Automation